

1 詳細リスク評価書シリーズ 20 亜鉛 第Ⅴ章 参考資料

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4 評価書の本文は「詳細リスク評価書シリーズ 20 亜鉛」（丸善株式会社）として 2008 年 5 月に

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藻類の急性毒性試験

生物種	エンドポイント	濃度 (mg/l)	温度 (°C)	文献
緑藻 <i>Selenastrum capricornutum</i>	24h EC50 (増殖)	0.015 (n)	24	Chen & Lin (1997)
	72h EC50 (増殖)	0.039 (m)	22	Muyssen & Janssen (2001a)
	72h EC50 (増殖)	0.117 (m)	22	
緑藻 <i>Chlorella vulgaris</i>	96h EC50 (増殖)	2.4 (n)	15.5	Rachlin & Farran (1974)
	72h EC50 (クロロフィル量)	2.4 (n)	25	Wren & McCarroll (1990)
	72h EC50 (増殖)	0.034 (m)	22	Muyssen & Janssen (2001a)
	72h EC50 (増殖)	0.105 (m)	22	
珪藻 <i>Navicula incerta</i>	9h EC50 (増殖)	10.0	25	Rachlin <i>et al.</i> (1983)
カナダ藻 <i>Elodea canadensis</i>	24h EC50 (光合成量)	8.1 (n)		Brown & Rattigan (1979)

微生物の毒性試験。h:時間。

生物種	エンドポイント	濃度(mg/l)	温度(°C)	pH	文献
アスピディスカ属の一種 <i>Aspidisca cicada</i>	24h LC50	2.4	20	7.3	Madoni <i>et al.</i> (1992)
	24h LC50	50.0	20	7.6	
<i>A. lynceus</i>	24h LC50	50.0	20	7.6	
繊毛虫類の一種 <i>Blepharisma americanum</i>	24h LC50	1.05	20	7.3	
ラッパムシ目の一種 <i>Colpidium campulum</i>	24h LC50	1.85	20	7.3	
ユープロテス属の一種 <i>Euplotes affinis</i>	24h LC50	3.10	20	7.3	
<i>E. patella</i>	24h LC50	50.0	20	7.3	
<i>E. sp.</i>	24h LC50	2.39	20		
ゾウリムシ <i>Paramecium caudatum</i>	24h LC50	2.50	20	7.3	
繊毛虫の一種 <i>Uronema nigricans</i>	24h LC50	< 0.170	20	7.3	Janssen <i>et al.</i> (1995)
<i>Colpoda cuculus</i>	168 EC50s	1.37			
キロドネラ属の一種 <i>Chilodonella uncinata</i>	24h LC50	2.9	20	7.6	Madoni <i>et al.</i> (1996)
オペキュラリア属の一種 <i>Opercularia coacratia</i>	24h LC50	10.3	20	7.6	
<i>Opercularia minima</i>	24h LC50	0.2	20	7.6	

無脊椎動物の急性毒性試験。h, d, w はそれぞれ時間、日、週を表す。濃度での (n) は設定濃度、(m) は実測濃度を表す。

生物種	齢・条件	エンドポイント	濃度 mg/l	硬度 CaCO ₃ (mg/L)	温度 ℃	pH	文献
Sludge Worm <i>Tubifex tubifex</i> イトミミズ		48h LC50	0.11 (n)	0.1	20	6.3	BrkovicPopovic & Popovic (1977)
		48h LC50	2.98 (n)	34.2	20	6.85	
		48h LC50	2.57 (n)	34.2	20	7.2	
		48h LC50	60.2 (n)	261	20	7.32	
		48h LC50	21.13 (n)	245	30	7.6	Khangarot (1991)
		96h LC50	17.78 (n)	245	30	7.6	
		48h LC50	20.81(m)	237	20	7.5	Rathore & Khangarot (2002)
		96h LC50	14.74(m)	237	20	7.5	
Oligochaete <i>Lumbriculus variegatus</i> オヨギミミズの一種		48h LC50	8.1(n)	30	20	7.8	Bailey & Liu (1980)
		96h LC50	6.3(n)	30	20	7.8	
Snail <i>Amnicola</i> sp. カタツムリの一種	eggs	96h LC50	20.2 (m)	50	17	7.6	Rehwoldt <i>et al.</i> (1973)
	adult	96h LC50	14 (m)	50	17	7.6	
Pond snail <i>Lymnaea luteola</i> カタツムリの一種		48h LC50	15.40 (m)	195	17.5	7.4	Khangarot & Ray (1987b)
		96h LC50	11 (m)	195	17.5	7.4	
		48h LC50	11.58 (m)	198	22	7.4	
		96h LC50	8.01 (m)	198	22	7.4	
		48h LC50	6.75 (m)	205	26	7.6	
		96h LC50	5 (m)	205	26	7.6	
		48h LC50	3.8 (m)	203	32	7.3	

		96h LC50	1.68 (m)	203	32	7.3	
Mollusc <i>Ancylus fluviatilis</i>	<2 mm	96h LC50	3.2 (n)		10		Willis (1988)
	>3 mm	96h LC50	4.5 (n)		10		
Annelid <i>Erpobdella oculata</i>	<4 mg	96h LC50	2.05 (n)		10		
	>15 mg	96h LC50	8.8 (n)		10		
Bristle worm <i>Nais</i> sp. 多毛虫の一種		96h LC50	18.4 (m)	50	17	7.6	Rehwoldt <i>et al.</i> (1973)
Water flea <i>Daphnia magna</i> オオミジンコ	餌無し	48h LC50	0.1 (n)	4453	1719	7.48.2	Biesinger & Christensen (1972)
	餌あり	48h LC50	0.28 (n)	4453	1719	7.48.2	
	<48 h	48h LC50	0.24 (n)	175		6	LeBlanc (1982)
	<24 h	48h LC50	0.151 (n)		20	6.5	Oikari <i>et al.</i> (1992)
	<24 h	48h LC50	0.244 (n)		20	6.5	
	<24 h	48h LC50	0.068 (n)	45		7.27.4	Mount & Norberg (1984)
	<24 h	48h LC50	0.75 (n)		20		Arambasic <i>et al.</i> (1995)
		48h LC50	1.22 (m)				Magliette <i>et al.</i> (1995)
	neonate	48h LC50	0.7989(m)	130	20	6.95	Attar & Maly (1982)
	<24 h	48h LC50	0.098(m)		20		Gale <i>et al.</i> (1992)
	Neonate Clone A	48h LC50	0.121(m)	46.1	20	7.2	Barata <i>et al.</i> (1998)
	Neonate Clone A	48h LC50	0.378(m)	90.7	20	7.73	
	Neonate Clone A	48h LC50	0.397(m)	179	20	8.07	
	Neonate Clone C	48h LC50	0.239(m)	46.1	20	7.2	
	Neonate	48h LC50	0.878(m)	90.7	20	7.73	

	Clone A						
	Neonate Clone A	48h LC50	0.636(m)	179	20	8.07	
	< 6 h	48h LC50	0.091 (m)	359	2325		Calson & Roush (1985)
	< 6 h	48h LC50	0.134 (m)	380	2325		
	< 6 h	48h LC50	0.124 (m)	274	2325		
	< 6 h	48h LC50	0.224 (m)	353	2325		
	< 6 h	48h LC50	0.114 (m)	376	2325		
	< 6 h	48h LC50	0.096 (m)	392	2325		
	< 6 h	48h LC50	0.264 (m)	362	2325		
	< 6 h	48h LC50	0.196 (m)	392	2325		
	< 6 h	48h LC50	0.041 (m)	45	2325		
	< 6 h	48h LC50	0.032 (m)	45	2325		
		48h LC50	0.56 (n)	240	13	7.6	
<i>D. pulex</i>	<24 h	48h LC50	0.107 (n)	45		7.27.4	Mount & Norberg (1984)
<i>C. dubia</i>	<48 h	48h LC50	>0.53(m)	300	25	6.06.5	SchubauerBerigan & Dierkes (1993)
	<48 h	48h LC50	0.36(m)	300	25	7.07.5	
	<48 h	48h LC50	0.096(m)	300	25	8.08.5	
		48h LC50	0.50 (m)		25		Magliette <i>et al.</i> (1995)
	neonate	48h LC50	0.07 (m)	97.6	25	6	Belanger & Cherry (1990)
		48h LC50	0.101 (m)	97.6	25	8	
		48h LC50	0.109 (m)	97.6	25	9	
		48h LC50	0.065 (m)	113.6	25	6	
		48h LC50	0.120 (m)	113.6	25	8	
		48h LC50	0.131 (m)	113.6	25	9	

		48h LC50	0.105 (m)	182	25	6	
		48h LC50	0.123 (m)	182	25	8	
		48h LC50	0.153 (m)	182	25		
	neonate	48h LC50	0.1993 (n)	7090	20		Diamond <i>et al.</i> (1997)
		7d LC50	0.1 (n)	6080	20	6.07.0	GillespieJr <i>et al.</i> (1999)
	<24 h	48h LC50	0.1227 (n)	81	25	7.74	Nelson & Roline (1998)
<i>C. reticulata</i>	<24 h	48h LC50	0.076 (n)	45		7.27.4	Mount & Norberg (1984)
	<6 h	48h LC50	0.032(m)	45	2325		Calson & Roush (1985)
<i>D. hyalina</i>	1. 27 mm	48h LC50	0.04 (n)		10	7.2	Baudouin & Scoppa (1974)
<i>D. lumholtzi</i>		48h LC50	2.29 (n)	200	28.5	7.9	Vardia <i>et al.</i> (1988)
		96h LC50	0.44 (n)	200	28.5	7.9	
<i>Moina irrasa</i>	<24 h	48h LC50	0.037 (n)	<5	20	5	Zou & Bu (1994)
		48h LC50	0.073 (n)	<5	20	6.5	
		96h LC50	0.049 (n)	<5	20	6.5	
		48h LC50	0.099 (n)	<5	20	8	
		48h LC50	0.024 (n)	<5	25	5	
		48h LC50	0.045 (n)	<5	25	6.5	
		48h LC50	0.059 (n)	<5	25	8	
<i>M. macrocopa</i> タマミジンコ		48h LC50	1.17 (n)		2427	6.5	Wong (1992)
Copepod <i>Cyclops abyssorum</i>	1. 27 mm	48h LC50	5.5 (n)		10	7.2	Baudouin & Scoppa (1974)
Copepod <i>Eudiaptomus padanus</i>	1. 27 mm	48h LC50	0.50 (n)		10	7.2	
Copepod <i>Parastenocaris germanica</i>	adult	48h LC50	4.5 (m)	10*	10.5	6.8	Notenboom <i>et al.</i> (1992)
	adult	96h LC50	1.7 (m)		10.5	6.8	

Copepod <i>Tropocyclops prasinus mexicanus</i>	adult	48h LC50	0.052 (n)	10	18.1	6.65	Lalande & PinelAllol (1986)
	adult	48h LC50	0.264 (n)	10	18.1	6.4	
	adult	48h LC50	2.934 (n)		18.1	7.1	
Amphipod <i>Gammarus</i> sp.		96h LC50	8.1 (m)	50	17	7.6	Rehwoldt <i>et al.</i> (1973)
Amphipod <i>Crangonyx pseudogracilis</i>	4 mm	48h LC50	121 (n)	50	13	6.75	Martin & Holdich (1986)
	4 mm	96h LC50	19.8 (n)	50	13	6.75	
Isopod <i>Asellus aquaticus</i>	7 mm	96h LC50	18.2 (n)	50	13	6.75	
Ostracod <i>Cypris subglobosa</i> 橈脚類の一種		48h LC50	34.99 (n)	200	28.5	7.9	Vardia <i>et al.</i> (1988)
		96h LC50	8.35 (n)	200	28.5	7.9	
Harpacticoid <i>Nitocra spinipes</i> ソコミジンコ目の 一種	adult	96h LC50	4.3 (n)	7	21	7.8	Linden <i>et al.</i> (1979)
Rotifer <i>Brachionus calyciflorus</i> ツボウムシ	juvenile	24h LC50	1.32 (n)	36.2	20	7.3	Couillard <i>et al.</i> (1989)
		24h LC50	1.3 (n)		25		Snell <i>et al.</i> (1991)
<i>Chironomus</i> sp. ユスリカの一種		96h LC50	18.2 (m)	50	17	7.6	Rehwoldt <i>et al.</i> (1973)
<i>C. tentans</i> ユスリカの一種	3rd instar	48h EC50	8.2 (n)	25	13	6.3	Khangarot & Ray (1989a)
Caddisfly Unidentified		96h LC50	58.1 (m)	50	17	7.6	Rehwoldt <i>et al.</i> (1973)
Damselfly Unidentified		96h LC50	26.2 (m)	50	17	7.6	
<i>Hyalella azteca</i> ヨコエビの一種	714 d	96h LC50	1.2 (m)	300	25	6.06.5	SchubauerBerigan & Dierkes (1993)
	714 d	96h LC50	1.5 (m)	300	25	7.07.5	
	714 d	96h LC50	0.29 (m)	300	25	8.08.5	
Crayfish <i>Orconectes virilis</i>	adult	2w LC50	84 (m)	26	18	7.1	Mirenda (1986)

<i>Ancylus fluviatilis</i> 貝	3mm > b1	96h LC50	4.5 (n)		10		Willis (1988)
	3mm > b1	100d LC50	0.13 (n)		10		
	2mm < b1	96h LC50	3.2 (n)		10		
	2mm < b1	100d LC50	0.08 (n)		10		
<i>Anodonta cygnea</i> ドブ貝		48h EC50 (蓋閉)	0.0691 (n)		13	8	Huebner & Pynnonen (1992)
Zebra mussel <i>Dreissena polymorpha</i> カワホトトギスガイ	1. 62. 0 cm	48h EC50 (filtration rate)	1.35(m)	150	15	7.9	Kraak <i>et al.</i> (1994a)
	1. 62. 2 cm	48h EC50 (filtration rate)	0.56 (m)	150	15	7.9	
	1. 62. 2cm	3w EC50 (fil- tration rate)	0.131 (m)	150	15	7.9	
貝の一種 <i>Anodonta imbecillis</i>	< 2d	48h LC50	0.355(m)	39	23		Keller & Zam (1991)
	< 2d	96h LC50	0.268(m)	39	23		
	< 2d	48h LC50	0.588(m)	45	23		
	< 2d	96h LC50	0.618(m)	45	23		
タイワンシジミ <i>Corbicula fluminea</i>	10-21mm	96h EC50 (閉殻)	6.04(m)			7.7	Rodgers <i>et al.</i> (1980)
サカマキガイの一種 <i>Physa gyrina</i>	adult	96h LC50	1.274(m)	36	15	6.9	Nebeker <i>et al.</i> (1986)
	adult	30d LC50	1.274(m)	36	15	6.9	

淡水魚類の急性毒性試験

生物種	齢・条件	エンド ポイント	濃度 (mg/l)	硬 度 (CaCO3 mg/l)	温度	pH	文献
Chinook salmon <i>Oncorhynchus tshawytscha</i>	fry	96h LC50	1.27 (n)	211	12	7.4-8.3	Hamilton & Buhl (1990)
	juvenile	96h LC50	0.084 (m)	20-21	11-13	7.17.2	Finlayson & Verrue (1982)
	alevin	96h LC50	>0.661 (m)	23	12	7.1	Chapman (1978b)
	swimup	96h LC50	0.097 (m)	23	12	7.1	
	parr	96h LC50	0.463 (m)	23	12	7.1	
	smolt	96h LC50	0.701 (m)	23	12	7.1	
Coho salmon <i>O. kisutch</i>	alevin	96h LC50	0.727 (n)	41.3	12	7.1-8.0	Buhl & Hamilton (1990)
	0.47 g	96h LC50	0.82 (n)	41.3	12	7.1-8.0	
	0.63 g	96h LC50	1.81 (n)	41.3	12	7.1-8.0	
	0.94 g	96h LC50	1.65 (n)	41.3	12	7.1-8.0	
	2.7 kg	96h LC50	0.905 (m)	25	14	7.4	Chapman & Stevens (1978)
Rainbow trout <i>O. mykiss</i>	alevin	96h LC50	2.17 (n)	41	12	7.1-8.0	Buhl & Hamilton (1990)
	0.60 g	96h LC50	0.169 (n)	41	12	7.1-8.0	
	juvenile	96h LC50	0.55 (m)			6.4-8.3	Hale (1977)
	alevin	96h LC50	0.815 (m)	23	12	7.1	Chapman (1978b)
	swimup	96h LC50	0.093 (m)	23	12	7.1	

	parr	96h LC50	0.136 (m)	23	12	7.1	
	smolt	96h LC50	>0.651 (m)	23	12	7.1	
	2.7 kg	96h LC50	1.76 (m)	83	10.3	7.45	Chapman & Stevens (1978)
	juvenile	96h LC50	0.43 (n)	26	15	6.8	Sinley <i>et al.</i> (1974)
	juvenile	96h LC50	7.21 (n)	333	15	7.81	
	25-70 g	96h LC50	2.6 (m)	137	12.7	7.3	Meisner & Quan Hum (1987)
	160-290 g	96h LC50	2.4 (m)	143.4	12.9	7.1	
	juvenile	96h LC50	0.37 (m)	46.8	14.9	7.63	Holcombe & Andrew (1978)
	juvenile	96h LC50	0.51 (m)	47	15.2	7.58	
	juvenile	96h LC50	0.756 (m)	44.4	15	7.38	
	juvenile	96h LC50	2.51 (m)	177.6	14.9	7.41	
	juvenile	96h LC50	2.96 (m)	179	15.7	7.17	
	juvenile	96h LC50	1.91 (m)	169.7	14.8	7.31	
	51-76mm	96h LC50	0.55 (n)			6.48.3	Hale (1977)
	juvenile	96h LC50	0.88 (m)	31.4	15	5.56	Bradley & Sprague (1985)
	juvenile	96h LC50	11.1 (m)	389	15	5.59	
	juvenile	96h LC50	9.95 (m)	394	15	5.46	
	juvenile	96h LC50	0.17 (m)	30.2	15	7	
	juvenile	96h LC50	0.19 (m)	31.2	15	7.04	
	juvenile	96h LC50	0.11 (m)	31.3	15	6.97	

	juvenile	96h LC50	4.46 (m)	387	15	6.99	
	juvenile	96h LC50	5.16 (m)	389	15	7.05	
	juvenile	96h LC50	4.53 (m)	30.9	15	9.01	
	juvenile	96h LC50	>87.9 (m)	373	15	8.97	
Sockeye salmon	parr	96h LC50	0.749 (m)	22	12	7.3	Chapman (1978a)
O. nerke	亜鉛濃度 0.24mg/l に 順化させた 仔魚	96h LC50	1.633 (m)	22	12	7.3	Chapman (1978a)
Cutthroat trout Salmo clarki	0.6 g	96h LC50	0.152	38	10	7.5	Mayer & Ellersieck (1986)
	0.9 g	96h LC50	0.6	43	15	7.5	
	0.9 g	96h LC50	0.13	40	10	7.8	
	1.0 g	96h LC50	0.061	40	10	8.5	
	1.0 g	96h LC50	0.1	38	10	6.5	
	1.0 g	96h LC50	0.074	38	5	7.5	
	1.59 g	96h LC50	0.09 (m)		14.3	6.67.6	Rabe & Sappington (1970)
Brook Trout <i>Salvelinus fontinalis</i>	juvenile	96h LC50	1.55 (m)	46.8	14.9	7.63	Holcombe <i>et al.</i> (1979)
	juvenile	96h LC50	2.12 (m)	47	15.2	7.58	
	juvenile	96h LC50	2.42 (m)	44.4	15	7.38	
	juvenile	96h LC50	6.14 (m)	177.6	14.9	7.41	

	juvenile	96h LC50	6.98 (m)	179	15.7	7.17	
	juvenile	96h LC50	4.98 (m)	169.7	14.8	7.31	
Brown Trout		14d LC50	0.64 (m)		8.6-12.0	7.3	Nehring & Goettl Jr. (1974)
Fathead minnow <i>Pimephale promelas</i>	79 mg	96h LC50	2.61 (n)	220	25	7.8	Broderius & Smith (1979)
	12 g	96h LC50	0.96 (n)	20	25	7.5	Pickering & Henderson (1966)
	12 g	96h LC50	0.78 (n)	20	25	7.5	
	12 g	96h LC50	33.4 (n)	360	25	8.2	
	12 g	96h LC50	0.88 (n)	20	25	7.5	
	12 g	96h LC50	0.77 (n)	20	25	7.5	
	12 g	96h LC50	2.33 (n)	20	15	7.5	
	12 g	96h LC50	2.55 (n)	20	15	7.5	
	immature	96h LC50	9.2 (m)	203		7.7	Brungs (1969)
	<24 h	96h LC50	0.238 (m)	48	25		Norberg & Mount (1985)
	juvenile	96h LC50	2.148 (m)	380	2325		Calson & Roush (1985)
	juvenile	96h LC50	2.159 (m)	362	2325		
	juvenile	96h LC50	2.0 (m)	374	2325		
	juvenile	96h LC50	0.396 (m)	45	2325		
	juvenile	96h LC50	0.6 (m)	46	25	7.08.0	Benoit & Holcombe (1978)
	<24 h	96h LC50	0.78 (m)	300	25	6.06.5	SchubauerBerigan & Dierkes

	<24 h	96h LC50	0.33 (m)	300	25	7.07.5	(1993)
	<24 h	96h LC50	0.5 (m)	300	25	8.08.5	
minnow Phoxinus phoxinus	adult	96h LC50	3.2 (m)	70	11.9	7.6	Bengtsson (1974)
Arctic grayling Thymallus arcticus	fry	96h LC50	0.315 (n)	41	12	7.1-8.0	Buhl & Hamilton (1990)
	alevin	96h LC50	1.58 (n)	41	12	7.1-8.0	
	alevin	96h LC50	2.92 (n)	41	12	7.1-8.0	
	0.20 g	96h LC50	0.142 (n)	41	12	7.1-8.0	
	0.34 g	96h LC50	0.112 (n)	41	12	7.1-8.0	
	0.85 g	96h LC50	0.166 (n)	41	12	7.1-8.0	
	1.85 g	96h LC50	0.47 (n)	41	12	7.1-8.0	
	1.85 g	96h LC50	0.168 (n)	41	12	7.1-8.0	
Bluegill Lepomis macrochirus	12 g	96h LC50	5.46 (n)	20	25	7.5	Pickering & Henderson (1966)
	12 g	96h LC50	4.85 (n)	20	25	7.5	
	12 g	96h LC50	5.82 (n)	20	25	7.5	
	12 g	96h LC50	40.9 (n)	360	25	8.2	
	12 g	96h LC50	5.37 (n)	20	25	7.5	
	12 g	96h LC50	5.46 (n)	20	25	7.5	
	12 g	96h LC50	6.44 (n)	20	15	7.5	

Killifish <i>Nothobranchius guenterei</i>		24h LC50	8.6	4044		7.8	Shedd <i>et al.</i> (1999)
Pumpkinseed	<20cm	96h LC50	20.0 (m)	53	17	7.8	Rehwoldt <i>et al.</i> (1971)
<i>Lepomis gibbosus</i>	<20cm	96h LC50	20.1 (m)	55	28	8	Rehwoldt <i>et al.</i> (1972)
Banded killifish	<20cm	96h LC50	19.1 (m)	53	17	7.8	Rehwoldt <i>et al.</i> (1971)
<i>Fundulus diaphanus</i>	<20cm	96h LC50	19.2 (m)	55	28	8	Rehwoldt <i>et al.</i> (1972)
<i>Roccus saxatilis</i>	<20cm	96h LC50	6.7 (m)	53	17	7.8	Rehwoldt <i>et al.</i> (1971)
	<20cm	96h LC50	6.8 (m)	55	28	8	Rehwoldt <i>et al.</i> (1972)
Striped bass	35-80 d	96h LC50	0.120 (n)	40	20	8.1	Palawski <i>et al.</i> (1985)
<i>Morone saxtilis</i>	35-80 d	96h LC50	0.430 (n)	285	20	7.9	
White perch	<20cm	96h LC50	14.3 (m)	53	17	7.8	Rehwoldt <i>et al.</i> (1971)
<i>Roccus americanus</i>	<20cm	96h LC50	14.4 (m)	55	28	8	Rehwoldt <i>et al.</i> (1972)
American eel	<20cm	96h LC50	14.6 (m)	53	17	7.8	Rehwoldt <i>et al.</i> (1971)
<i>Anguilla rostrata</i>	<20cm	96h LC50	14.5 (m)	55	28	8	Rehwoldt <i>et al.</i> (1972)
Japanese eel <i>Anguilla japonica</i>	77 g	96h LC50	11.0 (m)		25		Cruz & Morgan (1985)
Carp	<20cm	96h LC50	7.8 (m)	53	17	7.8	Rehwoldt <i>et al.</i> (1971)
<i>Cyprinus carpio</i>	<20cm	96h LC50	7.8 (m)	55	28	8	Rehwoldt <i>et al.</i> (1972)
	3.2 cm	96h LC50	0.45 (n)		15	7.1	Alam & Maughan (1992)

	3.2 cm	96h LC50	1.34 (n)		15	7.1	Khangarot <i>et al.</i> (1983) Rao <i>et al.</i> (1975)
	6.0 cm	96h LC50	1.64 (n)		15	7.1	
	6.0 cm	96h LC50	2.25 (n)		15	7.1	
	47-62 mm	96h LC50	3.12 (n)	19	15	6.3	
	45cm	96h LC50	0.15 (n)	108	25	7.5	
Northern Squawfish <i>Ptychocheilus oregonensis</i>	6.58 g	96h LC50	3.498 (m)	2030	9.5	7.17.5	Andros & Garton (1980)
	0.51 g	96h LC50	3.693 (m)	2030	12.4	7.17.5	
Colorado squawfish <i>Ptychocheilus lucius</i>	swimup fry	96h LC50	1.7 (n)	197	25	7.9	Hamilton (1995)
	47 mg	96h LC50	3.34 (n)	199	25	7.9	Buhl & Hamilton (1996)
	24-47 mg	96h LC50	8.4 (n)	144	25	8.06	Hamilton & Buhl (1997)
	0.41.1 g	96h LC50	4.3 (n)	197	25	7.9	Hamilton (1995)
	1.7 g	96h LC50	12 (n)	197	25	7.9	
	0.430.57 g	96h LC50	8.62 (n)	199	25	7.9	Buhl & Hamilton (1996)
Bonytail <i>Gila elegans</i>	swimup fry	96h LC50	4.8 (n)	197	25	7.9	Hamilton (1995)
	1.1 g	96h LC50	5.8 (n)	197	25	7.9	
	2.6 g	96h LC50	23 (n)	197	25	7.9	
	2-13 mg	96h LC50	5.35 (n)	199	25	7.9	Buhl & Hamilton (1996)
	0.38-0.45 g	96h LC50	8.01 (n)	199	25	7.9	

Razorback sucker ヌメリゴイ Xyrauchen texanus	swimup fry	96h LC50	4.1 (n)	197	25	7.9	Hamilton (1995)
	642 mg	96h LC50	9.8 (n)	144	25	8.06	Hamilton & Buhl (1997)
	0.9 g	96h LC50	6.5 (n)	197	25	7.9	Hamilton (1995)
	2.0 g	96h LC50	16 (n)	197	25	7.9	
	0.35-0.58 g	96h LC50	2.92 (n)	199	25	7.9	Buhl & Hamilton (1996)
Hawkfish Cirrhinus mrigala	fry	96h LC50	7.0 (m)	67	28.1	7.4	Sharma & Sharma (1995)
	fingerling	96h LC50	35.0 (m)	67	28.1	7.4	
	fingerling	96h LC50	1.4428 (n)	250-263	25.5-27.0	7.0-7.5	Gupta & Sharma (1994)
Goldfish Carassius auratus	12 g	96h LC50	6.44 (n)	20	25	7.5	Pickering & Henderson (1966)
Guppy Poecilia reticulata	0.10.2 g	96h LC50	1.27 (n)	20	25	7.5	
	5d	96h LC50	1.74 (m)	30 (EDTA)	25	7.16	Pierson (1980)
	male	96h LC50	300 (n)	118	22	8.3	Sehgal & Saxena (1986)
	female	96h LC50	278 (n)	118	22	8.3	
Gambusia affinis カダヤシ科	male	96h LC50	115	50		7.3	Kallanagoudar & Patil (1997)
	female	96h LC50	90	50		7.3	
	fry	96h LC50	50	50		7.3	
	male	96h LC50	140	150		7.3	
	female	96h LC50	120	150		7.3	

	fry	96h LC50	80	150		7.3	
	male	96h LC50	150	300		7.3	
	female	96h LC50	140	300		7.3	
	fry	96h LC50	100	300		7.3	
Flagfish Jordanella floridae	larvae	96h LC50	1.5 (n)	44	25	7.17.8	Spehar (1976)
Channelfish 学名なんやねん	500 mg	96h LC50	3.7 (n)	4		6.1	Abbasi & Soni (1986)
zebrafish Brachydanio rerio	embryo/larv ae	144- hNOEC(death)	>20 (n)	308.8	2627	7.6	Meinelt & Stueber (1995)
	embryo/larv ae	144- hNOEC(death)	1.5 (n)	62.5	2627	6.75	
Tilapia Tilapia zilli	subadult	96h LC50	33 (n)	20-22	9.3	6.7	Hilmy <i>et al.</i> (1987a)
	subadult	96h LC50	21 (n)	20-22	15.3	6.7	
	subadult	96h LC50	21 (n)	20-22	18.5	6.7	
	subadult	96h LC50	13 (n)	20-22	25	6.7	
Catfish Clarius lazera	subadult	96h LC50	52 (n)	20-22	9.3	6.7	
	subadult	96h LC50	40 (n)	20-22	15.3	6.7	
	subadult	96h LC50	38 (n)	20-22	18.5	6.7	
	subadult	96h LC50	26 (n)	20-22	25	6.7	

Ambassis Sp. タカサゴイシモチの一種	840 mg	9d LC50			25		Skidmore & Firth (1970)
Chequered Rainbow-fish Melanotaenia splendida inornata	28–50mm	96h LC50	6.2 (m)	16.5	27	7.2	Baker & Walden (1984)
	28–50mm	96h LC50	4.8 (m)	16.5	27	7.2	
Blackstriped Rainbow-fish M. nigrans	36–42 mm	96h LC50	13.9 (m)	16.5	27	7.2	
	36–42 mm	96h LC50	6.8 (m)	16.5	27	7.2	
Flyspecked Hardyhead Craterocephalus stercusmuscarum	22–30 mm	96h LC50	0.6 (m)	16.5	27	7.2	
Mottled sculpin	juvenile	96h LC50	0.156 (m)	48.6	12.2	7.38	Woodling <i>et al.</i> (2002)
Cottus Bairdi	juvenile	96h LC50	0.439 (m)	154	12.4	7.5	Brinkman & Woodling (2005)

海域の藻類の毒性試験

生物種	条件	エンドポイント	濃度 (mg/l)	温度	文献
珪藻 <i>Nitzshia closterium</i>		96h EC50(増殖)	0.075 (n)	21	Stauber & Florence (1989)
		96h EC50 (増殖)	0.065 (n)	21	Stauber & Florence (1990)
	富栄養培地	96h EC50 (増殖)	0.85 (n)	21	Stauber & Florence (1989)
		96h EC50 (増殖)	0.271(n)	15.5	Rosko & Rachlin (1975)
珪藻 <i>Asterionella japonica</i>		72h EC50 (増殖)	0.058 (n)	23	Fisher & Jones (1980)

海域の無脊椎動物の毒性試験

生物種	齢・条件	エンドポイント	濃度 (mg/l)	塩分濃度(‰)	温度(℃)	文献
Starfish	11.2 g	24h LC50	390(n)	20	20	Eisler & Hennekey (1977)
<i>Asterias forbesi</i>	11.2 g	96h LC50	39 (n)	20	20	
ヒトデ	11.2 g	168h LC50	2.3(n)	20	20	
Purple sea urchin	sperm	80m EC50	0.262(m)	30	8.2-8.4	Dinnel <i>et al.</i> (1989)
Strongylocentrotus	embryo	120h EC50	0.23(m)	30	8.2-8.4	
purpuratus	sperm	80m EC50	0.0041(n)		15	Phillips <i>et al.</i> (1998)
ムラサキウニ	embryo	96h EC50	0.00972(n)		15	
Green sea urchin <i>Strongylocentrotus</i>	sperm	80m EC50	0.383(m)	30	8.2-8.4	Dinnel <i>et al.</i> (1989)

<i>droebachiensis</i>						
Sand dollar <i>Dendraster excentricus</i> タコノマクラ	sperm	72h EC50	0.028 (m)	30	12.513.0	Dinnel <i>et al.</i> (1989)
American oyster <i>Crassostrea virginica</i>	embryo	48h LC50	0.31 (n)	25	26	Calabrese <i>et al.</i> (1973)
Mussel <i>Mytilus edulis</i> ムラサキイガイ		24h LC50	20.8 (n)	7	12	Hietanen & Kristoffersson (1988)
	opening re- sponse	24h EC50	1.35(n)	7	12	
	byssogenesis	24h EC50	0.64(n)	7	12	
<i>M. edulis planulatus</i> ムラサキイガイ		96h LC50	2.5 (m)	34	20.6	Ahsanullah (1976)
		96h LC50	3.6 (m)		17.6	
	0.32g	96h LC50	4.3(m)		18.1	
<i>Perna viridis</i> ミドリイガイ	34cm	24h LC50	3.2 (m)	25-27	2528	Yap <i>et al.</i> (2004)
Bay scallop <i>Argopecten irradians</i> アメリカイタヤガイ	juvenile	96h LC50	2.25 (n)	25	20	Nelson <i>et al.</i> (1988)
Surf clam <i>Spisula solidissima</i>	juvenile	96h LC50	2.95 (n)	25	20	

アメリカウバガイ						
Softshell clam	4.6 g	24h LC50	320 (n)	20	20	Eisler & Hennekey (1977)
<i>Mya arenaria</i>	4.6 g	96h LC50	7.7 (n)	20	20	
オオノガイ	4.6 g	168h LC50	3.1 (n)	20	20	
Hard Clam	embryo	48h LC50	0.166 (n)	26	25	Calabrese & Nelson (1974)
<i>Mercenaria mercenaria</i> ホンビノスガイ	larvae	810d LC50	0.1954(n)	24	25	Calabrese <i>et al.</i> (1977)
Red abalone	胚発生	48h EC50	0.068(n)	33-36	13.0-16.0	Hunt & Anderson (1989)
<i>Haliotis rufescens</i>	metamorphosis	9d EC50	0.05(n)	33-36	14.0-17.5	
アカネアワビ	adult	48h LC50	1400 (n)	25	5	
	adult	48h LC50	1200 (n)	35	5	
	adult	96h LC50	140 (n)	15	5	
	adult	96h LC50	700 (n)	25	5	
	adult	96h LC50	750 (n)	35	5	
	adult	48h LC50	1000 (n)	15	10	
	adult	48h LC50	2100 (n)	35	10	
	adult	96h LC50	210 (n)	15	10	
	adult	96h LC50	900 (n)	25	10	
	adult	96h LC50	950 (n)	35	10	
	adult	48h LC50	320 (n)	15	15	

	adult	48h LC ₅₀	1200 (n)	25	15	
	adult	48h LC ₅₀	950 (n)	35	15	
	adult	96h LC ₅₀	60 (n)	15	15	
	adult	96h LC ₅₀	180 (n)	25	15	
	adult	96h LC ₅₀	250 (n)	35	15	
Squid <i>Loligo opalescens</i> ヤリイカ	larvae	96h LC ₅₀	>1.92 (m)	30	8.6	Dinnel <i>et al.</i> (1989)
Eastern mud snail <i>Nassarius obsoletus</i>	0.4 g	24h LC ₅₀	150 (n)	20	20	Eisler & Hennekey (1977)
	0.4 g	96h LC ₅₀	50 (n)	20	20	
	0.4 g	168h LC ₅₀	7.4 (n)	20	20	
Amphipod <i>Allorchestes compressa</i>	0.06 g	96h LC ₅₀	0.58 (m)	34.5	20.5	Ahsanullah (1976)
		96h LC ₅₀	2.0 (m)	34.1	20.3	Ahsanullah <i>et al.</i> (1988)
Amphipod <i>Corophium insidiosum</i> ドロクダムシ	812mm	96h LC ₅₀	1.9 (n)		19.1	Reish (1993)
	812mm	96h LC ₅₀	12.5 (n)		19.1	
Corophium volutator ドロクダムシ科の一種	adult	48h LC ₅₀	14 (n)	5	5	Bryant <i>et al.</i> (1985)
	adult	48h LC ₅₀	20 (n)	10	5	
	adult	48h LC ₅₀	25 (n)	15	5	

	adult	48h LC ₅₀	46 (n)	25	5	
	adult	48h LC ₅₀	54 (n)	35	5	
	adult	96h LC ₅₀	1 (n)	5	5	
	adult	96h LC ₅₀	4.6 (n)	10	5	
	adult	96h LC ₅₀	6.5 (n)	15	5	
	adult	96h LC ₅₀	12 (n)	25	5	
	adult	96h LC ₅₀	16 (n)	35	5	
	adult	48h LC ₅₀	13 (n)	5	10	
	adult	48h LC ₅₀	12 (n)	10	10	
	adult	48h LC ₅₀	31 (n)	15	10	
	adult	48h LC ₅₀	46 (n)	25	10	
	adult	96h LC ₅₀	1.6 (n)	10	10	
	adult	96h LC ₅₀	8.5 (n)	15	10	
	adult	96h LC ₅₀	11 (n)	25	10	
	adult	96h LC ₅₀	15 (n)	35	10	
	adult	48h LC ₅₀	7 (n)	5	15	
	adult	48h LC ₅₀	17 (n)	10	15	
	adult	48h LC ₅₀	16 (n)	15	15	
	adult	48h LC ₅₀	17 (n)	25	15	
	adult	48h LC ₅₀	27 (n)	35	15	

	adult	96h LC ₅₀	1.1 (n)	5	15	
	adult	96h LC ₅₀	3.2 (n)	10	15	
Corophium volutator ドロクダムシ科の一種	adult	96h LC ₅₀	3.4 (n)	15	15	Bryant <i>et al.</i> (1985)
	adult	96h LC ₅₀	4.4 (n)	25	15	
	adult	96h LC ₅₀	3.6 (n)	35	15	
	adult	192h LC ₅₀	1.1 (n)	25	15	
Harpacticoid copepod <i>Nitocra spinipes</i> ソコミジンコ目の 一種		96h LC ₅₀	0.85 (n)			Bengtsson & Bergstrom (1987)
		96h LC ₅₀	1.3 (n)			
		96h LC ₅₀	2.4 (n)			
		96h LC ₅₀	2.8 (n)			
Sandworm <i>Nereis virens</i> ゴカイ類の一種	7.6 g	24h LC ₅₀	20 (n)	20	20	Eisler & Hennekey (1977)
	7.6 g	96h LC ₅₀	8.1 (n)	20	20	
	7.6 g	168h LC ₅₀	2.6 (n)	20	20	
Annelida <i>Neanthes vaalii</i>	0.33 g	96h LC ₅₀	5.5 (m)	34.2	18.7	Ahsanullah (1976)
Annelida <i>Neanthes arenaceodentat</i>	juvenile	96h LC ₅₀	0.9(n)	5		Reish <i>et al.</i> (1976)
	adult	28d LC ₅₀	1.8(n)	5		
	juvenile	96h LC ₅₀	0.9(n)	5		
	adult	28d LC ₅₀	1.4(n)	5		
Annelida	juvenile	96h LC ₅₀	1.7(n)	5		

Capitella capitata	adult	28d LC50	3.5(n)	5		
	adult	28d LC50	1.25(n)	5		
Dungeness crab Cancer magister	zoea	96h LC50	0.586 (m)	30	8.5	Dinnel <i>et al.</i> (1989)
Fiddler crab Uca annulipes	24-29 mm	96h LC50	31.9 (n)	25	29	Devi (1987)
	24-29 mm 汚染地で採取	96h LC50	77 (n)	25	29	
U. triangularis	24-29 mm	96h LC50	39.1 (n)	25	29	
	24-29 mm 汚染地で採取	96h LC50	66.4 (n)	25	29	
Hermit crab Pagurus longicarpus	0.5 g	24h LC50	12 (n)	20	20	Eisler & Hennekey (1977)
	0.5 g	96h LC50	0.4 (n)	20	20	
	0.5 g	168h LC50	0.2 (n)	20	20	
Grapsid crab <i>Paragrapsus</i> <i>quadridentatus</i>	1.44 g	96h LC50	11 (m)	34.2	19.6	Ahsanullah (1976)
	1.44 g	120h LC50	10.5 (m)	34.2	19.6	
	zoez	96h LC50	1.23 (m)	35	17	Ahsanullah & Arnott (1978)
	adult	96h LC50	11 (m)	35	17	
Portunus pelagicus	zoea	48h LC50	0.56 (n)	35	2527	Greenwood & Fielder (1983)
	zoea	48h LC50	0.65 (n)	35	2527	

	zoea	48h LC ₅₀	0.77 (n)	35	2527	
<i>P. sanguinolentus</i>	zoea	48h LC ₅₀	0.62 (n)	35	2527	
<i>Charybdis feriatus</i>	zoea	48h LC ₅₀	0.96 (n)	35	2527	
Copepod <i>Tisbe holothuriae</i>		48h LC ₅₀	0.62 (n)	38	22.5	Verriopoulos & Dimas (1988)
<i>Scutellidium</i> sp.	adult	24h LC ₅₀	1.09 (m)	34.4-35.5	17	Arnott & Ahsanullah (1979)
<i>Paracalanus parvus</i>	adult	24h LC ₅₀	1.38 (m)	34.4-35.5	17	
<i>Acartia simplex</i>	adult	24h LC ₅₀	1.86 (m)	34.4-35.5	17	
Shrimp <i>Mysidopsis bahia</i>	24 h	96h LC ₅₀	0.499 (m)	30	21	Lussier <i>et al.</i> (1985)
<i>Neomysis integer</i>	juvenile	96h LC ₅₀	0.54 (m)	5	20	Verslycke <i>et al.</i> (2003)
	juvenile	96h LC ₅₀	0.037 (m)	25	20	
Grass shrimp <i>Palaemonetes pugio</i>	juvenile	48h LC ₅₀	11.3 (m)	10	20	Burton & Fisher (1990)
<i>Palaemon</i> sp.	0.28 g	96h LC ₅₀	9.5 (m)	35.5	19.5	Ahsanullah (1976)
Crangon crangon <i>Callinassa australiensis</i>		4d LC ₅₀	10.2 (m)	34.137.5	1820	Ahsanullah <i>et al.</i> (1981)
		7d LC ₅₀	1.98 (m)	34.137.5	1820	
		10d LC ₅₀	1.54 (m)	34.137.5	1820	
		14d LC ₅₀	1.15 (m)	34.137.5	1820	
Mysid	juvenile	48h LC ₅₀	0.458 (m)	3436	1315.5	Martin <i>et al.</i> (1989)

Holmesimysis costata	juvenile	96h LC ₅₀	0.097 (m)	3440	1316	
	juvenile	7d LC ₅₀	0.046 (m)	3536	1516	
Praunus flexuosus		96h LC ₅₀	8 (n)	4.5	5	McLusky & Hagerman (1987)
		96h LC ₅₀	14 (n)	9	5	
		96h LC ₅₀	16 (n)	13.5	5	
		96h LC ₅₀	23 (n)	18	5	
		96h LC ₅₀	23 (n)	27	5	
		48h LC ₅₀	8 (n)	4.5	15	
		48h LC ₅₀	16 (n)	9	15	
		48h LC ₅₀	16 (n)	13.5	15	
		48h LC ₅₀	16 (n)	18	15	
		48h LC ₅₀	16 (n)	22.5	15	
		48h LC ₅₀	16 (n)	27	15	
		48h LC ₅₀	8 (n)	4.5	15	
		96h LC ₅₀	10 (n)	18	15	
		96h LC ₅₀	16 (n)	22.5	15	
		96h LC ₅₀	16 (n)	27	15	
Prawn	50-70 mm	48h LC ₅₀	4.0 (n)	15	27.5	Sivadasan <i>et al.</i> (1986)
Metapenaeus dobsoni	50-70 mm	72h LC ₅₀	1.95 (n)	15	27.5	
	50-70 mm	96h LC ₅₀	1.7 (n)	15	27.5	

	30-50 mm	48h LC ₅₀	3 (n)	15	27.5	
	30-50 mm	72h LC ₅₀	1.4 (n)	15	27.5	
	30-50 mm	96h LC ₅₀	0.84 (n)	15	27.5	

海域の魚類の急性毒性

生物種	齢・条件	エンドポイント	濃度 (mg/l)	塩分濃度(‰)	温度(℃)	文献
Chinook salmon <i>Oncorhynchus tshawytscha</i>	2.6 g	96h LC ₅₀	2.88 (n)	brackish	11-13	Hamilton & Buhl (1990)
Atheriniform fish Rivulus marmoratus マングローブ・キリー フィッシュ	0.03-0.1 g	96h LC ₅₀	119.3 (m)	14	26-27	Lin & Dunson (1993)
	0.03-0.1 g	96h LC ₅₀	176.6 (m)	14	26-27	
	0.02-0.1 g	96h LC ₅₀	129.5 (n)	14	26-27	
Mummichog Fundulus heteroclitus マミチョッグ	juvenile	48h LC ₅₀	96.5 (m)	10	20	Burton & Fisher (1990)
	1.3 g	24h LC ₅₀	125 (n)	20	20	Eisler & Hennekey (1977)
	1.3 g	96h LC ₅₀	60 (n)	20	20	
	1.3 g	168h LC ₅₀	52 (n)	20	20	
Grey mullet Chelon labrosus	0.87 g	48h LC ₅₀	36.9 (m)	34.6	12	Taylor <i>et al.</i> (1985)
	0.87 g	72h LC ₅₀	22.5 (m)	34.6	12	

グレイ モレット	0.87 g	96h LC ₅₀	21.5 (m)	34.6	12	
English sole <i>Parophrys vetulus</i> イギリスガレイ	larvae	96h LC ₅₀	14.5 (n)		12	Shenker & Cherr (1990)
Bleak	8 cm	96h LC ₅₀	32 (n)	7	10	Linden <i>et al.</i> (1979)
Alburnus alburnus	8 cm	96h LC ₅₀	41.9 (n)	7	10	
Sheephead minnow <i>Cyprinodon variegatus</i> シープヘッドミノ	7 d	96h LC ₅₀	11.3 (m)	18 g/kg	25	Moreau <i>et al.</i> (1999)
Cabazon <i>Scorpaenichthys marmoratus</i>	larvae	96h LC ₅₀	0.1914 (n)	30	8.6	Dinnel <i>et al.</i> (1989)
Striped bass <i>Morone saxatilis</i>	35-80 d old	96h LC ₅₀	0.430 (n)	1	20	Palawski <i>et al.</i> (1985)